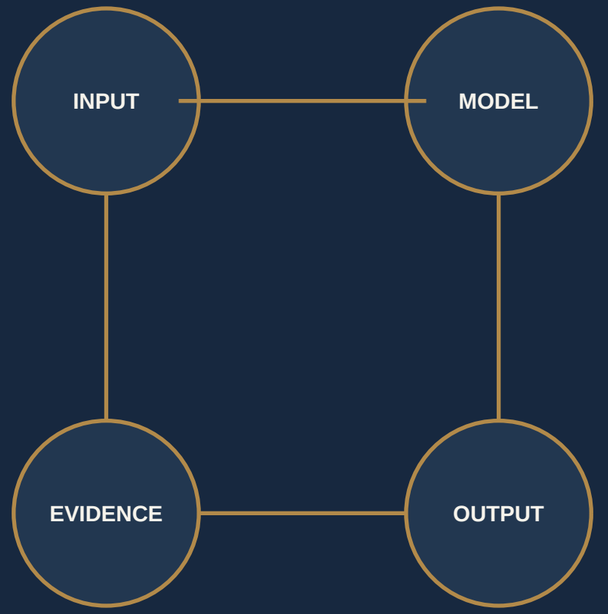
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Machine Vision

AI.307 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define industrial and robotic vision.
- Apply key methods including cameras.
- Build or analyze a practical system involving calibration.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

Industrial and robotic vision

02

METHODS

Cameras

03

SYSTEMS

Calibration

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · Industrial and robotic vision

Industrial and robotic vision

02 · Cameras

Cameras

03 · Calibration

Calibration

04 · Feature extraction

Feature extraction

05 · Object recognition

Object recognition

06 · Tracking

Tracking

07 · 3D vision and inspection

3D vision and inspection

08 · Mandatory practical: Build a m

Mandatory practical: Build a machine-vision inspection or robotic vision system

Core ideas, made explicit

AI.307 • Distinguish definitions, assumptions and evidence

- Industrial and robotic vision
- Cameras
- Calibration
- Feature extraction
- Object recognition

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

AI.307 • Distinguish definitions, assumptions and evidence

- Tracking
- 3D vision and inspection
- Mandatory practical: Build a machine-vision inspection or robotic vision system

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.

PROBLEM
Industrial and robotic vision

INPUTS
Cameras

METHOD
Calibration

OUTPUT
Feature extraction

EVALUATION
Evidence, constraints and limitations

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving industrial and robotic vision.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using industrial and robotic vision.
- Implement or simulate a baseline using cameras.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.